



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 2 - Day 2 Report

Internship Program — Batch 2026

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1. Introduction

This report summarizes the activities completed during the M-Tech Production & Marketing AI/ML Internship Program on **7th July 2026**. During today's session, I studied **Experiment 04** from the Artificial Intelligence & Machine Learning Lab Manual. This experiment focused on understanding the fundamentals of Machine Learning, its workflow, and the key concepts required to build and evaluate ML models. I also trained my first machine learning model using scikit-learn.

2. Objectives

- Define machine learning and understand the three main types of learning
 - Understand the end-to-end machine learning workflow step by step
 - Distinguish between features, labels, training data, and testing data
 - Understand the data-science project lifecycle
 - Train a first linear regression model using scikit-learn
 - Apply machine learning concepts to real-world problems
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3. Task Description

Today's work involved studying Experiment 04 from the AI/ML Lab Manual. The experiment introduced the fundamental concepts of Machine Learning, including how it differs from traditional programming, the three main types of ML (Supervised, Unsupervised, and Reinforcement Learning), and key terminology like features, labels, training, and testing. I also learned the seven-step ML workflow and the wider data-science project lifecycle. Finally, I trained my first machine learning model using Linear Regression to understand the relationship between hours studied and exam scores.

4. Work Done

During today's study session, I carefully completed Experiment 04 from the lab manual. I started by understanding what Machine Learning actually is and how it differs from traditional programming. I learned that in traditional programming we write rules and get answers, while in machine learning we provide data and answers and the computer learns the rules by itself.

I then studied the three main types of machine learning. I understood that Supervised Learning uses labeled data and includes both Regression (predicting numbers like house prices) and Classification (predicting categories like spam or not spam). I learned that Unsupervised Learning finds patterns in unlabeled data, like grouping customers into segments. I also learned about Reinforcement Learning where an agent learns through trial and error by receiving rewards or penalties.

Next, I studied the key terms in machine learning. I understood that features are input variables used to make predictions, labels are the outputs we want to predict, training is the process of learning patterns from data, and testing is checking the model on unseen data.

I then learned the seven-step ML workflow: collecting data, cleaning and preprocessing it, splitting into training and test sets, training a model, evaluating it, tuning hyperparameters, and finally deploying the model for real-world use. I also understood the broader data-science project lifecycle that includes business understanding, data acquisition, modeling, evaluation, deployment, and monitoring.

Finally, I trained my first machine learning model. I used Linear Regression to predict exam scores based on hours studied. I created a small dataset, trained the model, and made predictions. I also calculated the slope and intercept to understand what the model learned. This hands-on practice helped me see how ML models work in practice.

5. Challenges Faced

The concepts covered in Experiment 04 were well structured and beginner-friendly. However, understanding the difference between classification and regression took some time. I also had to spend extra time understanding the bias-variance tradeoff and why we need separate training and test sets. By reviewing the examples and practicing with different datasets, I gained a much clearer understanding of these concepts. The hands-on practice of training my own model was very helpful in reinforcing the theoretical concepts.

6. Key Learnings

Today's work significantly strengthened my understanding of machine learning fundamentals. I learned that machine learning is about learning patterns from data rather than being explicitly programmed. I understood the three main types of learning and their real-world applications. I learned the complete ML workflow from data collection to deployment. I also trained my first machine learning model using Linear Regression, which gave me practical experience in applying ML concepts. These concepts provide the essential building blocks for all future machine learning experiments.

7. Conclusion

Studying Experiment 04 provided a solid theoretical and practical foundation in Machine Learning fundamentals. I now understand what ML is, how it differs from traditional programming, and the complete workflow from data to deployment. Training my first Linear Regression model gave me hands-on experience and confidence to work with more complex ML algorithms in future experiments. This learning will serve as a strong base for upcoming experiments on supervised learning, unsupervised learning, and more advanced AI applications

throughout the internship.

Intern Signature:



Supervisor:

Name: Ghazi Muhammad Abdullah

Name:

Date: 07 July 2026

Designation: